

Question ID b52e5b6f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: b52e5b6f

PSAT2 M 1.1

A mixture consisting of only vitamin D and calcium has a total mass of **150** grams. The mass of vitamin D in the mixture is **50** grams. What is the mass, in grams, of calcium in the mixture?

- A. 200
- B. 150
- C. 100
- D. 50

ID: b52e5b6f Answer

Correct Answer: C

Rationale

Choice C is correct. Let d represent the mass, in grams, of vitamin D in the mixture, and let c represent the mass, in grams, of calcium in the mixture. It's given that the mixture consists of only vitamin D and calcium and that the total mass of the mixture is 150 grams. Therefore, the equation $d + c = 150$ represents this situation. It's also given that the mass of vitamin D in the mixture is 50 grams. Substituting 50 for d in the equation $d + c = 150$ yields $50 + c = 150$. Subtracting 50 from both sides of this equation yields $c = 100$. Therefore, the mass of calcium in the mixture is 100 grams.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the total mass, in grams, of the mixture, not the mass, in grams, of calcium in the mixture.

Choice D is incorrect. This is the mass, in grams, of vitamin D in the mixture, not the mass, in grams, of calcium in the mixture.

Question Difficulty: Easy

Question ID fa80893a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: fa80893a

PSAT2 M 1.2

If $2x + 3 = 9$, what is the value of $6x - 1$?

ID: fa80893a Answer

Correct Answer: 17

Rationale

The correct answer is 17. It’s given that $2x + 3 = 9$. Multiplying each side of this equation by 3 yields $3(2x + 3) = 3(9)$, or $6x + 9 = 27$. Subtracting 10 from each side of this equation yields $6x + 9 - 10 = 27 - 10$, or $6x - 1 = 17$. Therefore, the value of $6x - 1$ is 17.

Question Difficulty: Easy

Question ID a478f9f5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: a478f9f5

PSAT2 M 1.3

Each of **157** gemstones can be classified as one of three classifications, as shown in the frequency table.

Classification	Frequency
color X	119
color Y	3
color Z	35

If one of the gemstones is selected at random, what is the probability of selecting a gemstone of color Y?

- A. $\frac{3}{157}$
- B. $\frac{35}{157}$
- C. $\frac{119}{157}$
- D. $\frac{154}{157}$

ID: a478f9f5 Answer

Correct Answer: A

Rationale

Choice A is correct. If one of the gemstones is selected at random, the probability of selecting a gemstone of color Y is equal to the number of gemstones of color Y divided by the total number of gemstones. According to the table, there are 3 gemstones of color Y, and it's given that the total number of gemstones is 157. Therefore, if one of the gemstones is selected at random, the probability of selecting a gemstone of color Y is $\frac{3}{157}$.

Choice B is incorrect. This is the probability of selecting a gemstone of color Z.

Choice C is incorrect. This is the probability of selecting a gemstone of color X.

Choice D is incorrect. This is the probability of selecting a gemstone that's not of color Y.

Question Difficulty: Easy

Question ID ed856e9c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: ed856e9c

PSAT2 M 1.4

What is the y-intercept of the graph of $y = 34x + 81$ in the xy-plane?

- A. (0, 81)
- B. (0, 34)
- C. (0, −34)
- D. (0, −81)

ID: ed856e9c Answer

Correct Answer: A

Rationale

Choice A is correct. In the xy-plane, the graph of an equation in the form $y = mx + b$, where m and b are constants, has a slope of m and a y-intercept of $(0, b)$. Therefore, the y-intercept of the graph of $y = 34x + 81$ is $(0, 81)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

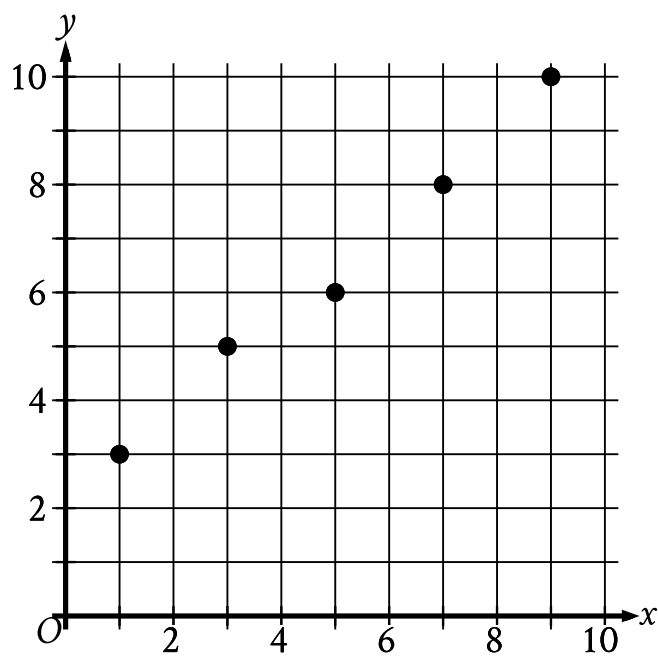
Question ID 16988f9c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 16988f9c

PSAT2 M 1.5

The scatterplot shows the relationship between two variables, x and y .



Which equation is the most appropriate linear model for this relationship?

- A. $y = -0.9x - 2.2$
- B. $y = -0.9x + 2.2$
- C. $y = -0.9x$
- D. $y = 0.9x + 2.2$

ID: 16988f9c Answer

Correct Answer: D

Rationale

Choice D is correct. A linear model can be written in the form $y = mx + b$, where m is the slope of the graph of the model in the xy -plane and $(0, b)$ is the y -intercept. The graph of an appropriate linear model for this relationship passes near the points $(1, 3)$ and $(9, 10)$ in the xy -plane. Two points on a line, x_1, y_1 and x_2, y_2 , can be used to find the slope of the line using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $(1, 3)$ and $(9, 10)$ for x_1, y_1 and x_2, y_2 , respectively, in the slope formula yields $m = \frac{10 - 3}{9 - 1}$, or $m = 0.875$. Therefore, the value of m for an appropriate linear model is approximately 0.875. Substituting 0.875 for m in $y = mx + b$ yields $y = 0.875x + b$. Since an appropriate linear model passes near the point $(1, 3)$, the approximate value of b can be found by substituting 1 for x and 3 for y in the equation $y = 0.875x + b$, which yields $3 = (0.875)(1) + b$, or $3 = 0.875 + b$. Subtracting 0.875 from both sides of this equation yields $2.125 = b$. Therefore, the value of b for an appropriate linear model is approximately 2.125. Thus, of the given choices, $y = 0.9x + 2.2$ is the most appropriate linear model for this relationship.

Alternate approach: A linear model can be written in the form $y = mx + b$, where m is the slope of the graph of the model in the xy -plane and $(0, b)$ is the y -intercept. The scatterplot shows that as the x -values of the data points increase, the y -values of the data points increase, which means the graph of an appropriate linear model has a positive slope. Of the given choices, $y = 0.9x + 2.2$ is the only linear model whose graph has a positive slope.

Choice A is incorrect. The graph of this model has a negative slope, not a positive slope.

Choice B is incorrect. The graph of this model has a negative slope, not a positive slope.

Choice C is incorrect. The graph of this model has a negative slope, not a positive slope.

Question Difficulty: Easy

Question ID 0700a2d5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 0700a2d5

PSAT2 M 1.6

How many yards are equivalent to 77 rods? (5.5 yards = 1 rod)

ID: 0700a2d5 Answer

Correct Answer: 423.5, 847/2

Rationale

The correct answer is 423.5. It's given that 5.5 yards = 1 rod. Therefore, 77 rods is equivalent to $77 \text{ rods} \times \frac{5.5 \text{ yards}}{1 \text{ rod}}$, or 423.5 yards. Note that 423.5 and 847/2 are examples of ways to enter a correct answer.

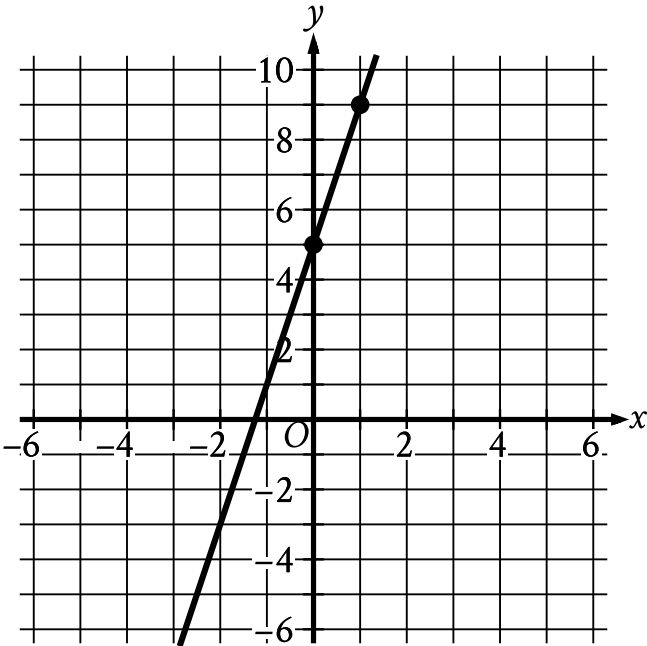
Question Difficulty: Easy

Question ID b8fa27db

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: b8fa27db

PSAT2 M 1.7



Line j is shown in the xy -plane. Line k (not shown) is parallel to line j . What is the slope of line k ?

ID: b8fa27db Answer

Correct Answer: 4

Rationale

The correct answer is 4. It's given that line k is parallel to line j . It follows that the slope of line k is equal to the slope of line j . Given two points on a line in the xy -plane, x_1, y_1 and x_2, y_2 , the slope of the line can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. In the xy -plane shown, the points $0, 5$ and $1, 9$ are on line j . It follows that the slope of line j is $\frac{9 - 5}{1 - 0}$, or 4. Since the slope of line j is equal to the slope of line k , the slope of line k is also 4.

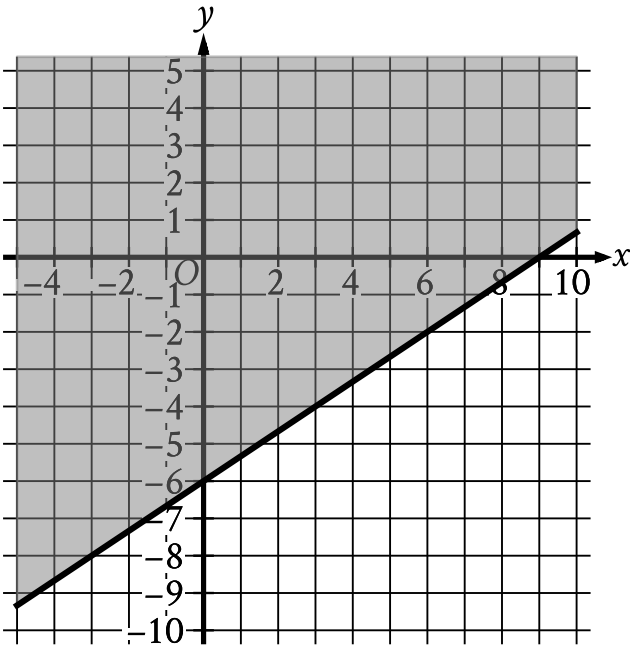
Question Difficulty: Easy

Question ID c9355dec

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: c9355dec

PSAT2 M 1.8



The shaded region shown represents the solutions to which inequality?

- A. $y \geq \frac{2}{3}x - 6$
- B. $y \geq \frac{2}{3}x + 6$
- C. $y \geq \frac{2}{3}x - 9$
- D. $y \geq \frac{2}{3}x + 9$

ID: c9355dec Answer

Correct Answer: A

Rationale

Choice A is correct. The equation for the line representing the boundary of the shaded region can be written in slope-intercept form $y = mx + b$, where m is the slope and b is the y -intercept of the line. For the graph shown, the boundary line passes through the points $(0, -6)$ and $(9, 0)$. Given two points on a line, (x_1, y_1) and (x_2, y_2) , the slope of the line can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $(0, -6)$ and $(9, 0)$ for (x_1, y_1) and (x_2, y_2) , respectively, in this equation yields $m = \frac{0 - (-6)}{9 - 0}$, which is equivalent to $m = \frac{6}{9}$, or $m = \frac{2}{3}$. Since the point $(0, -6)$ represents the y -intercept, it follows that $b = -6$. Substituting $\frac{2}{3}$ for m and -6 for b in the equation $y = mx + b$ yields $y = \frac{2}{3}x - 6$ as the equation of the boundary line. Since the shaded region represents all the points on and above this boundary line, it follows that the shaded region shown represents the solutions to the inequality $y \geq \frac{2}{3}x - 6$.

Choice B is incorrect. This inequality represents a region whose boundary line has a y -intercept of $(0, 6)$, not $(0, -6)$.

Choice C is incorrect. This inequality represents a region whose boundary line has a y -intercept of $(0, -9)$, not $(0, -6)$.

Choice D is incorrect. This inequality represents a region whose boundary line has a y -intercept of $(0, 9)$, not $(0, -6)$.

Question Difficulty: Easy

Question ID e86f0651

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: e86f0651

PSAT2 M 1.9

A circle has a radius of **43** meters. What is the area, in square meters, of the circle?

- A. $\frac{43\pi}{2}$
- B. 43π
- C. 86π
- D. $1,849\pi$

ID: e86f0651 Answer

Correct Answer: D

Rationale

Choice D is correct. The area, A , of a circle is given by the formula $A = \pi r^2$, where r is the radius of the circle. It's given that the circle has a radius of 43 meters. Substituting 43 for r in the formula $A = \pi r^2$ yields $A = \pi 43^2$, or $A = 1,849\pi$. Therefore, the area, in square meters, of the circle is $1,849\pi$.

Choice A is incorrect. This is the area, in square meters, of a circle with a radius of $\sqrt{\frac{43}{2}}$ meters.

Choice B is incorrect. This is the area, in square meters, of a circle with a radius of $\sqrt{43}$ meters.

Choice C is incorrect. This is the circumference, in meters, of the circle.

Question Difficulty: Medium

Question ID 3a3b95df

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 3a3b95df

PSAT2 M 1.10

$d = 16 - \frac{x}{30}$

The equation shown gives the estimated amount of diesel d , in gallons, that remains in the gas tank of a truck after being driven x miles, where $0 \leq x \leq 480$. What is the estimated amount of diesel, in gallons, that remains in the gas tank of the truck when $x = 300$?

- A. 0
- B. 6
- C. 14
- D. 16

ID: 3a3b95df Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the equation $d = 16 - \frac{x}{30}$ gives the estimated amount of diesel d , in gallons, that remains in the gas tank of the truck after being driven x miles. Substituting 300 for x in the given equation yields $d = 16 - \frac{300}{30}$, which is equivalent to $d = 16 - 10$, or $d = 6$. Therefore, the estimated amount of diesel that remains in the gas tank of the truck when $x = 300$ is 6 gallons.

Choice A is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 480$, not when $x = 300$.

Choice C is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 60$, not when $x = 300$.

Choice D is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 0$, not when $x = 300$.

Question Difficulty: Easy

Question ID 739f1bbc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 739f1bbc

PSAT2 M 1.11

In triangle ABC , $AB = 4,680$ millimeters (mm) and $BC = 4,680$ mm. Which statement is sufficient to prove that triangle ABC is equilateral?

- A. $AC = 4,680$ mm
- B. $AC = 468$ mm
- C. $AC = 46.8$ mm
- D. $AC = 4.68$ mm

ID: 739f1bbc Answer

Correct Answer: A

Rationale

Choice A is correct. In an equilateral triangle, all three sides have the same length. It’s given that in triangle ABC , $AB = 4,680$ mm and $BC = 4,680$ mm. Therefore, if $AC = 4,680$ mm, then all three sides of triangle ABC have the same length, so triangle ABC is equilateral. Therefore, $AC = 4,680$ mm is sufficient to prove that triangle ABC is equilateral.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 07bcecac

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 07bcecac

PSAT2 M 1.12

$$P(t) = 24.8(1.036)^t$$

The function P gives the predicted population, in millions, of a certain country for the period from 1984 to 2018, where t is the number of years after 1984. According to the model, what is the best interpretation of the statement “ $P(8)$ is approximately equal to 32.91”?

- A. In 1984, the predicted population of this country was approximately 8 million.
- B. In 1984, the predicted population of this country was approximately 32.91 million.
- C. 8 years after 1984, the predicted population of this country was approximately 32.91 million.
- D. 32.91 years after 1984, the predicted population of this country was approximately 8 million.

ID: 07bcecac Answer

Correct Answer: C

Rationale

Choice C is correct. The function P gives the predicted population, in millions, of a certain country for the period from 1984 to 2018, where t is the number of years after 1984. Since the value of $P8$ is the value of Pt when $t = 8$, it follows that “ $P8$ is approximately equal to 32.91” means that the value of Pt is approximately equal to 32.91 when $t = 8$. Therefore, the best interpretation of the statement “ $P8$ is approximately equal to 32.91” is that 8 years after 1984, the predicted population of this country was approximately 32.91 million.

Choice A is incorrect. In 1984, the predicted population of this country was 24.8 million, not approximately 8 million.

Choice B is incorrect. In 1984, the predicted population of this country was 24.8 million, not approximately 32.91 million.

Choice D is incorrect. 32.91 years after 1984, the predicted population of this country was $24.8(1.036)^{32.91}$ million, or approximately 79.42 million, not approximately 8 million.

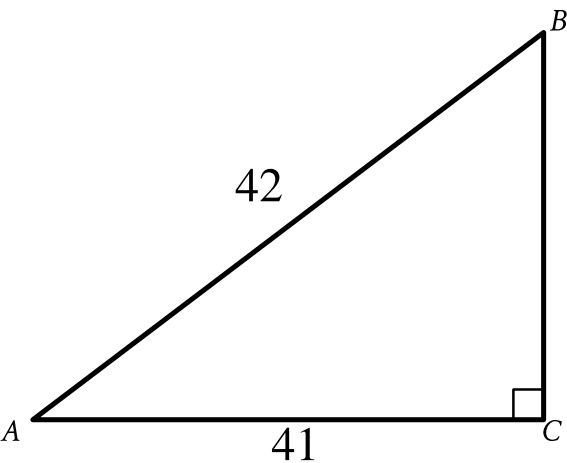
Question Difficulty: Easy

Question ID 2bddbc1b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 2bddbc1b

PSAT2 M 1.13



Note: Figure not drawn to scale.

What is the value of $\cos A$ in the triangle shown?

- A. $\frac{42}{41}$
- B. $\frac{41}{42}$
- C. $\frac{1}{42}$
- D. $\frac{1}{41}$

ID: 2bddbc1b Answer

Correct Answer: B

Rationale

Choice B is correct. The cosine of an acute angle in a right triangle is defined as the ratio of the length of the leg adjacent to the angle to the length of the hypotenuse. In the triangle shown, the length of the leg adjacent to angle A is 41, and the length of the hypotenuse is 42. Therefore, $\cos A = \frac{41}{42}$.

Choice A is incorrect. This is the value of $\frac{1}{\cos A}$. Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 79cf8505

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 79cf8505

PSAT2 M 1.14

A line segment that has a length of **115 centimeters (cm)** is divided into three parts. One part is **47 cm** long. The other two parts have lengths that are equal to each other. What is the length, in **cm**, of one of the other two parts of equal length?

ID: 79cf8505 Answer

Correct Answer: 34

Rationale

The correct answer is 34. It’s given that a line segment has a length of 115 cm and is divided into three parts, where one part is 47 cm long and the other two parts have lengths that are equal. If x represents the length, in cm, of each of the two parts of equal length, then the equation $47 + x + x = 115$, or $47 + 2x = 115$, represents this situation. Subtracting 47 from each side of this equation yields $2x = 68$. Dividing each side of this equation by 2 yields $x = 34$. Therefore, the length, in cm, of one of the two parts of equal length is 34.

Question Difficulty: Medium

Question ID a29e89fc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: a29e89fc

PSAT2 M 1.15

The list gives the mass, in grams, of 5 alpine marmots. 4,010; 4,010; 3,030; 4,050; 3,050
What is the mean mass, in grams, of these 5 alpine marmots?

ID: a29e89fc Answer

Correct Answer: 3630

Rationale

The correct answer is 3,630. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. The sum of the masses, in grams, of these alpine marmots is $4,010 + 4,010 + 3,030 + 4,050 + 3,050$, or 18,150 grams. The number of alpine marmots in the data set is 5. Therefore, the mean mass, in grams, of these 5 alpine marmots is $\frac{18,150}{5}$, or 3,630.

Question Difficulty: Medium

Question ID b4acba95

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: b4acba95

PSAT2 M 1.16

$x^2 - 12x + 27 = 0$ How many distinct real solutions does the given equation have?

- A. Exactly two
- B. Exactly one
- C. Zero
- D. Infinitely many

ID: b4acba95 Answer

Correct Answer: A

Rationale

Choice A is correct. The number of solutions of a quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. If the value of the discriminant is positive, then the quadratic equation has exactly two distinct real solutions. If the value of the discriminant is equal to zero, then the quadratic equation has exactly one real solution. If the value of the discriminant is negative, then the quadratic equation has zero real solutions. In the given equation, $x^2 - 12x + 27 = 0$, $a = 1$, $b = -12$, and $c = 27$. Substituting these values for a , b , and c in $b^2 - 4ac$ yields $-12^2 - 4(1)(27)$, or $-144 - 108$, or -252 . Since the value of its discriminant is negative, the given equation has zero distinct real solutions.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 946ab892

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 946ab892

PSAT2 M 1.17

For the linear function g , the graph of $y = g(x)$ in the xy -plane has a slope of 2 and passes through the point $(1, 14)$. Which equation defines g ?

- A. $g(x) = 2x$
- B. $g(x) = 2x + 2$
- C. $g(x) = 2x + 12$
- D. $g(x) = 2x + 14$

ID: 946ab892 Answer

Correct Answer: C

Rationale

Choice C is correct. An equation defining a linear function can be written in the form $gx = mx + b$, where m is the slope and $0, b$ is the y -intercept of the graph of $y = gx$ in the xy -plane. It's given that the graph of $y = gx$ has a slope of 2 . Therefore, $m = 2$. It's also given that the graph of $y = gx$ passes through the point $1, 14$. It follows that when $x = 1$, $gx = 14$. Substituting 1 for x , 14 for gx , and 2 for m in the equation $gx = mx + b$ yields $14 = 2(1) + b$, or $14 = 2 + b$. Subtracting 2 from each side of this equation yields $12 = b$. Therefore, $b = 12$. Substituting 2 for m and 12 for b in the equation $gx = mx + b$ yields $gx = 2x + 12$. Therefore, the equation that defines g is $gx = 2x + 12$.

Choice A is incorrect. For this function, the graph of $y = gx$ in the xy -plane passes through the point $1, 2$, not $1, 14$.

Choice B is incorrect. For this function, the graph of $y = gx$ in the xy -plane passes through the point $1, 4$, not $1, 14$.

Choice D is incorrect. For this function, the graph of $y = gx$ in the xy -plane passes through the point $1, 16$, not $1, 14$.

Question Difficulty: Medium

Question ID 1b403590

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 1b403590

PSAT2 M 1.18

An object has a mass of **168** grams and a volume of **24** cubic centimeters. What is the density, in grams per cubic centimeter, of the object?

- A. **7**
- B. **144**
- C. **192**
- D. **4,032**

ID: 1b403590 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the object has a mass of 168 grams and a volume of 24 cubic centimeters. Dividing the mass, in grams, of the object by the volume, in cubic centimeters, of the object gives the density, in grams per cubic centimeter, of the object. It follows that the density of the object is $\frac{168 \text{ grams}}{24 \text{ cubic centimeters}}$, which is equivalent to $\frac{168}{24}$ grams per cubic centimeter, or 7 grams per cubic centimeter.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 02add2d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 02add2d2

PSAT2 M 1.19

A company has a newsletter. In January 2018, there were 1,300 customers subscribed to the newsletter. For the next 24 months after January 2018, the total number of customers subscribed to the newsletter each month was 7% greater than the total number subscribed the previous month. Which equation gives the total number of customers, c , subscribed to the company's newsletter m months after January 2018, where $m \leq 24$?

- A. $c = 1,300^{msup}$
- B. $c = 1,300^{msup}$
- C. $c = 1,300^{msup}$
- D. $c = 1,300^{msup}$

ID: 02add2d2 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that in January 2018, there were 1,300 customers subscribed to a company's newsletter and for the next 24 months after January 2018, the total number of customers subscribed to the newsletter each month was 7% greater than the total number subscribed the previous month. It follows that this situation can be represented by the equation $c = a1 + \frac{r}{100}^m$, where c is the total number of customers subscribed to the company's newsletter m months after January 2018, a is the number of customers subscribed to the newsletter in January 2018, and the total number of customers subscribed to the newsletter each month was $r\%$ greater than the total number subscribed the previous month. Substituting 1,300 for a and 7 for r in this equation yields $c = 1,3001 + \frac{7}{100}^m$, or $c = 1,3001.07^m$.

Choice A is incorrect. This equation represents a situation where the total number of customers subscribed each month was 93% less, not 7% greater, than the total number subscribed the previous month.

Choice C is incorrect. This equation represents a situation where the total number of customers subscribed each month was 70%, not 7%, greater than the total number subscribed the previous month.

Choice D is incorrect. This equation represents a situation where the total number of customers subscribed each month was 600%, not 7%, greater than the total number subscribed the previous month.

Question Difficulty: Medium

Question ID 39714777

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 39714777

PSAT2 M 1.20

$p(x) + 57 = x^2$

The given equation relates the value of x and its corresponding value of $p(x)$ for the function p . What is the minimum value of the function p ?

- A. $-3,249$
- B. -57
- C. 57
- D. $3,249$

ID: 39714777 Answer

Correct Answer: B

Rationale

Choice B is correct. For a quadratic function defined by an equation of the form $px = ax - h^2 + k$, where a , h , and k are constants and $a > 0$, the minimum value of the function is k . Subtracting 57 from both sides of the given equation yields $px = x^2 - 57$. This function is in the form $px = ax - h^2 + k$, where $a = 1$, $h = 0$, and $k = -57$. Therefore, the minimum value of the function p is -57 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 1baffbcf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	■ ■ ■

ID: 1baffbcf

PSAT2 M 1.21

The expression $0.35x$ represents the result of decreasing a positive quantity x by what percent?

- A. 3.5%
- B. 35%
- C. 6.5%
- D. 65%

ID: 1baffbcf Answer

Correct Answer: D

Rationale

Choice D is correct. Let $n\%$ represent the percent by which the positive quantity x is decreased to result in $0.35x$. The value of n can be found by solving the equation $x - \frac{n}{100}x = 0.35x$. Since x is a common factor of each of the terms on the left-hand side of this equation, the equation can be rewritten as $x(1 - \frac{n}{100}) = 0.35x$. Dividing each side of this equation by x yields $1 - \frac{n}{100} = 0.35$. Multiplying each side of this equation by 100 yields $100 - n = 35$. Subtracting 100 from each side of this equation yields $-n = -65$. Dividing each side of this equation by -1 yields $n = 65$. Therefore, the expression $0.35x$ represents the result of decreasing the positive quantity x by 65%.

Choice A is incorrect. Decreasing the quantity x by 3.5% yields $x - 0.035x$, or $0.965x$, not $0.35x$.

Choice B is incorrect. Decreasing the quantity x by 35% yields $x - 0.35x$, or $0.65x$, not $0.35x$.

Choice C is incorrect. Decreasing the quantity x by 6.5% yields $x - 0.065x$, or $0.935x$, not $0.35x$.

Question Difficulty: Hard

Question ID 7866a908

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	■ ■ ■

ID: 7866a908

PSAT2 M 1.22

A sample of a certain alloy has a total mass of **50.0** grams and is **50.0%** silicon by mass. The sample was created by combining two pieces of different alloys. The first piece was **30.0%** silicon by mass and the second piece was **80.0%** silicon by mass. What was the mass, in grams, of the silicon in the second piece?

- A. **9.0**
- B. **16.0**
- C. **20.0**
- D. **30.0**

ID: 7866a908 Answer

Correct Answer: B

Rationale

Choice B is correct. Let x represent the total mass, in grams, of the first piece, and let y represent the total mass, in grams, of the second piece. It's given that the sample has a total mass of 50.0 grams. Therefore, the equation $x + y = 50.0$ represents this situation. It's also given that the sample is 50.0% silicon by mass. Therefore, the total mass of the silicon in the sample is 0.500×50.0 , or 25.0, grams. It's also given that the first piece was 30.0% silicon by mass and the second piece was 80.0% silicon by mass. Therefore, the masses, in grams, of the silicon in the first and second pieces can be represented by the expressions $0.300x$ and $0.800y$, respectively. Since the sample was created by combining the first and second pieces, and the total mass of the silicon in the sample is 25.0 grams, the equation $0.300x + 0.800y = 25.0$ represents this situation. Subtracting y from both sides of the equation $x + y = 50.0$ yields $x = 50.0 - y$. Substituting $50.0 - y$ for x in the equation $0.300x + 0.800y = 25.0$ yields $0.300(50.0 - y) + 0.800y = 25.0$. Distributing 0.300 on the left-hand side of this equation yields $15.0 - 0.300y + 0.800y = 25.0$. Combining like terms on the left-hand side of this equation yields $15.0 + 0.500y = 25.0$. Subtracting 15.0 from both sides of this equation yields $0.500y = 10.0$. Dividing both sides of this equation by 0.500 yields $y = 20.0$. Substituting 20.0 for y in the expression representing the mass, in grams, of the silicon in the second piece, $0.800y$, yields $0.800(20.0)$, or 16.0. Therefore, the mass, in grams, of the silicon in the second piece is 16.0.

Choice A is incorrect. This is the mass, in grams, of the silicon in the first piece, not the second piece.

Choice C is incorrect. This is the total mass, in grams, of the second piece, not the mass, in grams, of the silicon in the second piece.

Choice D is incorrect. This is the total mass, in grams, of the first piece, not the mass, in grams, of the silicon in the second piece.

Question Difficulty: Hard